Arbitrarily Long Runs of Large Values in OEIS A085710

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Abstract

OEIS A085710 records the smallest positive integer k such that $kn+1$ is semiprime. We prove that every term is defined, and for any positive integers K and L there are infinitely many blocks of L consecutive indices on which the sequence exceeds K. The first assertion follows from Dirichlet's theorem; the second has an elementary Chinese remainder theorem proof. The result does *not* settle the separate OEIS conjecture that $a(n) < n$ for all $n > 2$.

Keywords: semiprimes; prime factorization; Chinese remainder theorem; Dirichlet's theorem; OEIS A085710.

1. Definition and statement

Write $\Omega(m)$ for the number of prime factors of m, counted with multiplicity. A semiprime is an integer $m > 1$ with $\Omega(m) = 2$; prime squares are included. The OEIS entry A085710 defines

$$a(n) = \min\{k \geq 1 : \Omega(kn+1) = 2\} \qquad (n \geq 1), \tag{1}$$

provided that the set is nonempty [1].

Theorem 1. *For every $n \geq 1$, the value $a(n)$ is defined. Moreover, for every $K, L \geq 1$ there exist infinitely many positive integers N such that*

$$a(N+j) > K \qquad (0 \leq j < L). \tag{2}$$

Consequently, $\limsup_{n\to\infty} a(n) = +\infty$.

2. Every term is defined

Proof of the first assertion. For a fixed integer $n \geq 2$, Dirichlet's theorem gives infinitely many primes congruent to 1 modulo n. Fix one such prime p, and let q range over other primes in the same residue class. Then

$$pq \equiv 1 \pmod{n}, \qquad k = \frac{pq-1}{n} \in \mathbb{Z}_{>0}.$$

Because $p \neq q$, the integer $kn+1 = pq$ is a squarefree semiprime. Different q give infinitely many eligible k. When $n = 1$, the same argument works with any two distinct primes. The nonempty set of eligible positive k has a least element; hence (1) is defined. □

3. Simultaneous obstruction to small multipliers

Proof of the second assertion. Fix positive integers K, L. For each pair of indices

$$(j,k) \in \{0,\dots,L-1\} \times \{1,\dots,K\},$$

choose three primes $\ell_{j,k,1}, \ell_{j,k,2}, \ell_{j,k,3}$, all distinct over every triple of indices, and all greater than K. The integers k are therefore invertible modulo these primes. Impose

$$N \equiv -j - k^{-1} \pmod{\ell_{j,k,t}} \qquad (t = 1,2,3), \tag{3}$$

where k^{-1} is the multiplicative inverse modulo the indicated prime. The moduli are pairwise coprime, so the Chinese remainder theorem gives one residue class $N \equiv N_0 \pmod{M}$, with

$$M = \prod_{j=0}^{L-1} \prod_{k=1}^{K} \prod_{t=1}^{3} \ell_{j,k,t}.$$

For every such N and every (j,k) in the stated ranges, multiply (3) by k and add $kj+1$ to obtain

$$k(N+j)+1 \equiv 0 \pmod{\ell_{j,k,t}} \qquad (t = 1,2,3).$$

Thus $k(N+j)+1$ has at least three *distinct* prime divisors and cannot be a semiprime. For each fixed j this rules out all multipliers $1 \le k \le K$ in (1). By the first assertion, the least eligible multiplier nevertheless exists, so $a(N+j) > K$. There are infinitely many positive representatives $N = N_0 + hM$; therefore (2) holds for infinitely many N. Letting $L = 1$ and allowing K to grow proves the unboundedness. □

4. Scope, priority and AI disclosure

The OEIS entry conjectures the much stronger upper bound $a(n) < n$ for $n > 2$ [1]. Our construction makes $a(n)$ large for selected indices but supplies no counterexample to, or proof of, that bound. The arguments use standard theorems and no publication priority is claimed. The proof was generated and subjected to internal consistency and finite-case checks using AI; it has *not* been independently refereed or formally certified.

References

[1] The OEIS Foundation Inc., *A085710: Smallest k such that kn + 1 is a semiprime*, `https://oeis.org/A085710`, consulted 11 November 2026.

[2] Dirichlet's theorem on primes in arithmetic progressions (classical result).

[3] Chinese remainder theorem (classical result).